

SHIVENDRA SINGH

9918859850 singhshivendra1945@gmail.com [LinkedIn](#) [GitHub](#) [Portfolio](#)

Education

PDPM IITDM Jabalpur

Bachelor of Technology in Electronics and Communication Engineering — **CPI: 8.4** **Aug. 2024 – July 2028** *Jabalpur, Madhya Pradesh*

Central Hindu Boys School

Intermediate — **91%** **2022 – 2023** *Varanasi, Uttar Pradesh*

Technical Skills

Languages: C++, Python, JavaScript

CS Fundamentals: Data Structures and Algorithms, Object-Oriented Programming, Competitive Programming

Backend & APIs: Node.js, Express.js, Python (FastAPI), REST APIs, JWT, Authentication & Authorization

Databases: SQL, MySQL, SQLite, MongoDB, Mongoose, Sequelize ORM

Frontend: React.js, Redux Toolkit, React Router, HTML, CSS, Tailwind CSS

ML & Data: Scikit-learn, NumPy, Pandas, Matplotlib

Tools: Git, GitHub, Postman, VS Code

Projects

Quick Teams – Talent Matching & Workspace Platform

(React.js, Redux Toolkit, Node.js, Express.js, Python, FastAPI, SQL, JWT, Tailwind CSS) **Sept. 2026**

- Architected a full-stack developer workspace platform using **React.js/Redux Toolkit**, **Node.js/Express REST APIs**, and a **Python (FastAPI)** microservice to automate hackathon team formation.
- Engineered a Python-based multi-term skill tokenization and scoring algorithm to match project needs with candidate techstacks, integrating a dual-mode **Sequelize SQL** database.
- Implemented secure **JWT authentication**, multi-admin role-based access control (**RBAC**), and team workspaces featuring real-time group chat and invite workflows.

VideoTube – Backend & REST API Development

(Node.js, Express.js, MongoDB, Mongoose, JWT, Bcrypt, Multer, Cloudinary, REST APIs) **June 2026**

- Developed a modular RESTful backend with **40+ API endpoints** across **8+ functional modules** using Node.js, Express.js, MongoDB, and Mongoose.
- Implemented secure **authentication and authorization** using JWT access/refresh tokens, bcrypt password hashing, HTTP-only cookies, and protected middleware routes.
- Designed a reusable **MVC architecture** with modular controllers, routes, middleware, and utility components, leveraging MongoDB aggregation pipelines for data-intensive operations.

Radar-Based Object Detection & Classification System

(ESP32, CDM324 Radar Sensor, FFT, Python, Scikit-learn, CNN) **Jan. 2026 – Apr. 2026**

- Processed real-time radar signals using **FFT and spectrogram analysis**, extracting **20+ micro-Doppler features** for object classification.
- Optimized SVM classification logic, achieving a **15% improvement** in distinguishing between similar object classes under varied environmental conditions.

Achievements

- Achieved **Pupil** on [Codeforces](#) with a maximum rating of **1316**.
- Achieved **3-star** rating on [CodeChef](#) with a maximum rating of **1612**.
- Achieved a maximum rating of **1608** on [LeetCode](#).
- Solved **740+ DSA and Competitive Programming problems** across [different platforms](#).
- Made **700+ contributions** on [GitHub](#).

Leadership & Extracurricular

- Conducted technical sessions for **100+ students** in the college robotics club and led teams in **5+ technical competitions**.